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Introduction to Linux

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Objective

- Introduction to Linux Operating System.
- An overview of Linux Commands.

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What is Linux?

- **Unix** is a family of multitasking, multiuser computer operating systems.
- Derived from the original AT&T Unix, development starting in the 1970s at the Bell Labs research center by Ken Thompson, Dennis Ritchie, and others.
- **Linux**, is a family of open-source Unix-like operating systems based on the Linux kernel.
- An operating system kernel first released on September 17, 1991, by Linus Torvalds.

Why Unix/Linux ?

- Conceptually simple
- Consistent
- Extendible
- Modular

Linux Shell

- Computers understand the language of zeros and ones known as binary language.
- In the early days of computing, instructions were provided using binary language, which is difficult for all of us to read and write.
- Therefore, in an operating system there is a special program called the shell.
- The shell accepts human readable commands and translates them into something the kernel can read and process.

What is a Shell?

- The shell is a user program or it is an environment provided for user interaction.
- It is a command language interpreter that executes commands read from the standard input device such as keyboard or from a file.
- The shell gets started when you log in or open a console (terminal).
- Quick and dirty way to execute utilities.
- The shell is not part of system kernel, but uses the system kernel to execute programs, create files etc.

Types of shells for Linux

1. BASH (Bourne-Again SHell)
   - Most common shell in Linux. It's Open Source.
2. CSH (C SHell) –
   - The C shell's syntax and usage are very similar to the C programming language.
1. KSH (Korn SHell)
   - Created by David Korn at AT & T Bell Labs.
   - The Korn Shell also was the base for the POSIX Shell standard specifications.
2. TCSH
   - It is an enhanced but completely compatible version of the Berkeley UNIX C shell (CSH).

Shells Prompt

There are various ways to get shell access

- Terminal - Linux desktop provide a GUI based login system. Once logged in you can gain access to a shell byrunning X Terminal (XTerm), Gnome Terminal (GTerm), or KDE Terminal (KTerm) application.
- Connect via secure shell (SSH) - You will get a shell prompt as soon as you log in into remote server or workstation.
- Use the console - A few Linux system also provides a text-based login system. Generally you get a shell prompt as soon as you log in to the system.

To Find Your Current Shell Name

- To find all of the available shells in your system, type the following command:

```
mithun@LAPTOP-G8OP3754:~$ cat /etc/shells
# /etc/shells: valid login shells
/bin/sh
/bin/bash
/usr/bin/bash
/bin/rbash
/usr/bin/rbash
/bin/dash
/usr/bin/dash
/usr/bin/tmux
/usr/bin/screen
```

Command Line Interface (CLI)

- The shell provides an interface to Linux where you can type or enter commands using the keyboard.
- It is known as the command line interface (CLI).
- To find out your current shell type following command:

```
mithun@LAPTOP-G8OP3754:~$ echo $SHELL
/bin/bash
mithun@LAPTOP-G8OP3754:~$ ps $$
  PID TTY      STAT   TIME COMMAND
    7 tty1     S      0:00 -bash
mithun@LAPTOP-G8OP3754:~$ ps -p $$
  PID TTY          TIME CMD
    7 tty1     00:00:00 bash
```

What are Commands?

- A command is an instruction given to our computer by us to do whatever we want.
- In Mac OS, and Linux it is called terminal, whereas, in windows it is called command prompt.
- Commands are always case sensitive.
- Commands are executed by typing in at the command line followed by pressing enter key.
- This command further passes to the shell which reads the command and execute it.
- Shell is a method for the user to interact with the system.
- Default shell in Linux is called BASH = Bourne Again SHell.

Basic shell commands

- Command structure
  - Each command in Linux is made of a single word, with no spaces
  - Virtually every command is in all lowercase letters (Linux is case-sensitive)
- Entering commands
  - Entered at the prompt
  - The format of each command is:

    <command> [<option(s)>] [<argument(s)>]

    the items in square brackets are not always required.

Basic shell commands – Cont'd..

- Command may be
  - Name of an internal command (built into shell)
  - Name of a Linux utility
  - Name of a user-defined command

- Option part of the command
  - Exactly one character long
  - Each option preceded by the character –
  - May be followed by an argument for the option
  - Options with no argument can be grouped after a single –character
  - All options precede the command arguments
  - – may be used to indicate the end of options

Basic shell commands – Cont'd..

- Arguments
  - Specific to the command
  - Could be file names
- Command may result into an error if it is not properly parsed.
- Special key sequences
  - Used to perform certain functions from the shell

Key	Name	Action
^? or ^H	erase	Backup one space and delete character
^U	kill	Erase entire command from command line
^C	intr	interrupt/stop a currently executing command and return to prompt
^Z	susp	Suspend the currently running command It can later be restarted or killed

Basic shell commands – Cont'd..

- Return value
  - Every command returns a value to indicate if it executed successfully
  - Typically, return value 0 indicates success
  - Available in bash shell variable $?

Summary

- In this session we discussed an introduction to Linux operating system.
- And an overview of Linux commands.

Reference

- Linux Shell Scripting Tutorial - A Beginner's handbook: About, by Vivek Gite.

Objective

- Familiarize Linux Commands –(Part I)
  - Basic Linux Commands
  - Linux Directory Commands

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Basic Linux Commands

- man
- whatis
- echo
- whoami

man pages

- The "man" is a short term for manual page.
- Unix/Linux provides an online reference manual
  - Same as the printed reference manual
  - Sections of man page:

Section	Description
1	User commands
2	System calls
3	Subroutines
4	Devices (Special files and hardware)
5	File formats and configuration files
6	Games and demos
7	Miscellaneous: characters sets, filesystem types, etc
8	System administration and maintenance
l	Local
n	New

  - The command to get help on a given section is man (for manual).

man command

- **Syntax :** **man [option(s)] keyword(s)**
- **Example :** To Display the manual page for the item (program) ls.

  man ls

```
[root@localhost demo]# man ls
LS(1)                          User Commands                          LS(1)

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List  information  about  the FILEs (the current directory by default).
       Sort entries alphabetically if none of -cftuvSUX nor --sort  is  speci-
       fied.

       Mandatory  arguments  to  long  options are mandatory for short options
       too.
```

- **Output:** it will display man page of ls command.

whatis

- **whatis** command in Linux is used to get a one-line manual page descriptions.
- **Example :** To see just the description of a manual page, use ***whatis*** followed by a string.
  - **whatis man**

```
deepa@blackhawk:~$ whatis man
man (1)              - an interface to the on-line reference manuals
man (7)              - macros to format man pages
```

echo

- 'echo' displays some string.
- Example:
  - **echo "Hello World!"**

```
deepa@blackhawk:~$ echo "Hello World!"
Hello World!
```

whoami

- 'whoami' will print the user name from wherever you type the command.
- Example:
  - **whoami**

```
deepa@blackhawk:~$ whoami
deepa
```

Linux Directory Commands

- pwd
- cd
- ls
- mkdir
- rmdir

pwd

- **pwd** –"print working directory"
- This command displays your location currently you are working on.
- It will give the whole path starting from the root ending to the directory.
- Example:

```
deepa@blackhawk:~$ pwd
/home/deepa
```

cd

- cd means 'change directory'
- Linux **cd** command is used to change the current working directory (i.e., in which the current user is working).
- It is one of the most frequently used commands in the Linux terminal.
- Syntax:

cd <dirname>

Various cd operations on directories

1. Change from the current directory to a new directory
2. Change directory using an absolute path
3. Change directory using a relative path
4. Change to the home directory
5. Change to the previous directory
6. Change to Parent Directory

1. Change from the current directory to a new directory

- We can change our directory from the current working directory to a specified directory.
- To change our current working directory, execute the command as follows:

  cd < current directory> <specified directory>

```
deepa@blackhawk:~$ pwd
/home/deepa
deepa@blackhawk:~$ cd /home/deepa/Documents
deepa@blackhawk:~/Documents$
```

2.Change directory using an absolute path

- To change the directory by using an absolute path, we have to mention the whole path starting from the root.
- Example:

```
deepa@blackhawk:~$ cd /home/deepa/Documents/Test
deepa@blackhawk:~/Documents/Test$ cd /home/deepa/Documents/Test/Course
deepa@blackhawk:~/Documents/Test/Course$
```

3. Change directory using a relative Path

We can change our directory by using a relative path; a relative path is a location that is relative to the current directory.

- Example:

```
deepa@blackhawk:~/Documents$ cd Test
deepa@blackhawk:~/Documents/Test$
```

4. Change to the home directory

- To change the directory to home directory from the current working directory, execute the command as follows: cd ~

```
deepa@blackhawk:~$ cd /home/deepa/Documents/Test
deepa@blackhawk:~/Documents/Test$ cd ~
deepa@blackhawk:~$
```

5. Change to the previous directory

- To change to the previous directory from the current working directory, execute the command as follows: **cd -**

```
deepa@blackhawk:~/Documents/Test/Course$
deepa@blackhawk:~/Documents/Test/Course$ cd -
/home/deepa/Documents/Test
deepa@blackhawk:~/Documents/Test$
```

6. Change to Parent Directory

- To change the directory to parent directory of the current working directory, execute the command as follows: **cd ..**

```
deepa@blackhawk:~/Documents/Test$ cd ..
deepa@blackhawk:~/Documents$
```

ls

- ls' stands for 'list' command in Linux.
- It displays the contents of your current working directory.
- Example:

```
deepa@blackhawk:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
deepa@blackhawk:~$ cd /home/deepa/Documents/Test/Course
deepa@blackhawk:~/Documents/Test/Course$ ls
Courseplan  Syllabus
deepa@blackhawk:~/Documents/Test/Course$
```

ls -l

The ls command will only display the files. But if you want your files to be displayed in a long list format, then you can use ls -l command.

ls command options

```
deepa@blackhawk:~$ ls -l
total 32
drwxr-xr-x 2 deepa deepa 4096 Aug 29 14:33 Desktop
drwxr-xr-x 3 deepa deepa 4096 Jan  1 20:53 Documents
drwxr-xr-x 2 deepa deepa 4096 Aug 29 14:33 Downloads
drwxr-xr-x 2 deepa deepa 4096 Aug 29 14:33 Music
drwxr-xr-x 2 deepa deepa 4096 Jan  1 21:03 Pictures
drwxr-xr-x 2 deepa deepa 4096 Aug 29 14:33 Public
drwxr-xr-x 2 deepa deepa 4096 Aug 29 14:33 Templates
drwxr-xr-x 2 deepa deepa 4096 Aug 29 14:33 Videos
```

Columns above indicate specific things:

- Column 1 indicates information regarding file permission.
- Column 2 indicates the number of links to the file.
- Column 3 & 4 indicates the owner and group information.
- Column 5 indicates size of the file in bytes.
- Column 6 shows the date and time on which the file was recently modified.
- Column 7 shows the file or directory name.

ls command options

ls -a

It will give you the whole list of a directory including the hidden files also. In Linux, hidden files start with a dot (.) and can't be seen in the regular directory.

```
deepa@blackhawk:~$ ls -a
.               .bashrc  Documents      .local    .profile
..              .cache   Downloads      .mozilla  Public
.bash_history   .config  .gnupg         Music     Templates
.bash_logout    Desktop  .ICEauthority  Pictures  Videos
```

mkdir

- mkdir - 'make directory'.
- With the help of mkdir command, you can create a new directory wherever you want in your system.
- Syntax: mkdir <dirname>

```
deepa@blackhawk:~$ cd Documents/Test/
deepa@blackhawk:~/Documents/Test$ mkdir Exam
deepa@blackhawk:~/Documents/Test$ ls
Course  Exam
deepa@blackhawk:~/Documents/Test$
```

To make multiple directories

mkdir <dirname1> <dirname2> <dirname3> ...

```
deepa@blackhawk:~/Documents/Test$ mkdir Marks StudentList
deepa@blackhawk:~/Documents/Test$ ls
Course  Exam  Marks  StudentList
deepa@blackhawk:~/Documents/Test$
```

mkdir

Option	Description
mkdir –p , -parents	Add directory including its subdirectory.
mkdir –v, -verbose	Print a message for each created directory.

```
deepa@blackhawk:~/Documents/Test$ mkdir -p Sample/Work
deepa@blackhawk:~/Documents/Test$ ls
Course  Exam  Marks  Sample  StudentList
deepa@blackhawk:~/Documents/Test$ cd Sample/
deepa@blackhawk:~/Documents/Test/Sample$ ls
Work
deepa@blackhawk:~/Documents/Test/Sample$
```

```
deepa@blackhawk:~/Documents/Test$ mkdir -v Example
mkdir: created directory 'Example'
deepa@blackhawk:~/Documents/Test$ ls
Course  Exam  Example  Marks  Sample  StudentList
deepa@blackhawk:~/Documents/Test$
```

rmdir

- This command is used to delete a directory.
- But will not be able to delete a directory including a sub-directory.
- It means, a directory has to be empty to be deleted.
- Syntax: rmdir <dirname>

```
deepa@blackhawk:~/Documents/Test$ rmdir Sample
rmdir: failed to remove 'Sample': Directory not empty
deepa@blackhawk:~/Documents/Test$ cd Sample
deepa@blackhawk:~/Documents/Test/Sample$ ls
Work
deepa@blackhawk:~/Documents/Test/Sample$ rmdir Work
deepa@blackhawk:~/Documents/Test/Sample$ ls
deepa@blackhawk:~/Documents/Test/Sample$
```

Summary

- In this session we discussed Linux basic commands, and Linux directory commands.

Reference

- Cobbaut P. Linux Fundamentals. Samurai Media Limited; 2016.

Objective

- Familiarize Linux Commands –(Part II)
  - Working with Files
  - Working with File Content

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Linux Commands : Working with Files

- file
- touch
- rm
- cp
- mv
- rename

Create File

- Linux files are case sensitive.
- Example:
  - **test.txt** and **Test.txt** will be considered as two different files.

```
deepa@blackhawk:~/Documents/Test$ ls
Course  Exam  Example  Marks  Sample  StudentList  test.txt  Test.txt
deepa@blackhawk:~/Documents/Test$
```

- There are multiple ways to create a file in Linux.
  - using cat command
  - using touch command
  - using a different text editor like vim, nano, vi

1. Using cat command

- The cat command is one of the most used commands in Linux.
- It is used to **create a file, display the content of the file, concatenate the contents of multiple files, display the line numbers,** and more.
- **Syntax**: cat > *file_name.txt*
- To display the content of the file, execute the cat command as follows: *cat file_name.txt*

```
deepa@blackhawk:~/Documents/Test$ cat > datafile.txt
This is a text file.
deepa@blackhawk:~/Documents/Test$ cat datafile.txt
This is a text file.
deepa@blackhawk:~/Documents/Test$
```

2. Using the touch command

- It is used to create a new file, update the time stamp on existing files and directories.
- It creates empty files in Linux.
- To create a file, execute the touch command followed by the file name as given below:

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
ubuntu@ubuntu-Inspiron-3558:~/test$ touch f1
ubuntu@ubuntu-Inspiron-3558:~/test$ touch f2
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2
ubuntu@ubuntu-Inspiron-3558:~/test$
```

3. **Using Text Editor**

- We can also create a file using the different text editors like **vim, nano, vi,** and more.
  - **Using Vim text editor**
- To create a file using the vim text editor, execute the below command: vim test1.txt
- The text editor looks like as follows:

file

- file command is used to determine the file type.
- It does not care about the extension used for file.
- It simply uses file command and tell us the file type.
- Syntax: *file <filename>*

```
deepa@blackhawk:~/Documents/Test$ file data.txt
data.txt: empty
deepa@blackhawk:~/Documents/Test$ file test1.txt
test1.txt: ASCII text
deepa@blackhawk:~/Documents/Test$
```

rm

- The 'rm' means remove.
- This command is used to remove a file.
- The command line doesn't have a recycle bin or trash unlike other GUI's to recover the files.
- Once you have deleted a file, it is removed permanently.
- **Syntax:** rm <filename>

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2
ubuntu@ubuntu-Inspiron-3558:~/test$ rm f2
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1
ubuntu@ubuntu-Inspiron-3558:~/test$
```

rm - options

- rm –r : To delete a directory recursively.
- But one needs to be careful. Linux won't ask you to reconfirm your command. So, with any command, especially rm –r , one has to be careful!
- rm * : Used to delete files having same extension.
- This command removes all the files ending with same extensions like .pdf, .txt, .odt, .png, etc.

Example: rm -r

```
mithun@LAPTOP-G8OP3754:~$ cd exam
mithun@LAPTOP-G8OP3754:~/exam$ ls
shell.sh
mithun@LAPTOP-G8OP3754:~/exam$ cd
mithun@LAPTOP-G8OP3754:~$ rm -r exam
mithun@LAPTOP-G8OP3754:~$ cd exam
-bash: cd: exam: No such file or directory
```

Example: rm *

```
mithun@LAPTOP-G8OP3754:~$ ls
exam.txt  f1  file1.txt  folder2  google  pingmanual  script6.sh  shell.sh
ssh_keys  students.txt  study
mithun@LAPTOP-G8OP3754:~$ rm *.txt
mithun@LAPTOP-G8OP3754:~$ ls
f1  folder2  google  pingmanual  script6.sh  shell.sh  ssh_keys  study
```

cp

- 'cp' means copy. 'cp' command is used to copy a file or a directory.
- Syntax: cp <existing file name> <new file name>

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2
ubuntu@ubuntu-Inspiron-3558:~/test$ cp f1 f3
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2  f3
ubuntu@ubuntu-Inspiron-3558:~/test$
```

- If the target is a directory, then the source files are copied to that target directory.

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls test2
ubuntu@ubuntu-Inspiron-3558:~/test$ cp f1 test2
ubuntu@ubuntu-Inspiron-3558:~/test$ ls test2
f1
ubuntu@ubuntu-Inspiron-3558:~/test$
```

cp -r

- To copy complete directories, use cp -r (the -r option forces recursive copying of all files in all subdirectories).

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2  f3  test2
ubuntu@ubuntu-Inspiron-3558:~/test$ cp -r test2 test3
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2  f3  test2  test3
ubuntu@ubuntu-Inspiron-3558:~/test$
```

- Copy multiple files to directory
  - You can also use cp to copy multiple files into a directory. In this case, the last argument must be a directory.

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls test3
ubuntu@ubuntu-Inspiron-3558:~/test$ cp f1 f2 f3 test3
ubuntu@ubuntu-Inspiron-3558:~/test$ ls test3
f1  f2  f3
ubuntu@ubuntu-Inspiron-3558:~/test$
```

mv

- mv command is used to move existing file or directory from one location to another.
- It is also used to rename a file or directory.
- If you want to rename a single directory or file then **'mv'** option will be better to use.

Rename a file

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f1  f2  f3  test2  test3
ubuntu@ubuntu-Inspiron-3558:~/test$ mv f1 f4
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f2  f3  f4  test2  test3
ubuntu@ubuntu-Inspiron-3558:~/test$
```

Rename a directory

```
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f3  f4  test2  test3
ubuntu@ubuntu-Inspiron-3558:~/test$ mv test2 test4
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f3  f4  test3  test4
ubuntu@ubuntu-Inspiron-3558:~/test$
```

Linux Commands: Working with file contents

- head
- tail
- cat
- tac
- grep
- nl

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head

- The 'head' command displays the starting content of a file.
- By default, it displays starting 10 lines of any file.

- **Syntax:**
  head <file name>

```
ubuntu@ubuntu-Inspiron-3558:~/test$ head f3
one
two
three
four
five
six
seven
eight
nine
ten
ubuntu@ubuntu-Inspiron-3558:~/test$
```

- The head command can also display the first n lines of a file.

```
ubuntu@ubuntu-Inspiron-3558:~/test$ head -4 f3
one
two
three
four
ubuntu@ubuntu-Inspiron-3558:~/test$
```

tail

- tail command is used to display the last ten lines of one or more files.
- Its main purpose is to read the error message.
- By default, it displays the last ten lines of a file.
- Additionally, it is used to monitor the file changes in real-time.

```
ubuntu@ubuntu-Inspiron-3558:~/test$ tail f3
six
seven
eight
nine
ten
eleven
twelve
thirteen
fourteen
fifteen
ubuntu@ubuntu-Inspiron-3558:~/test$
```

cat

- The 'cat' command can be used to display the content of a file.

```
ubuntu@ubuntu-Inspiron-3558:~/test$ cat f3
one
two
three
four
five
six
seven
eight
nine
ten
eleven
twelve
thirteen
fourteen
fifteen
ubuntu@ubuntu-Inspiron-3558:~/test$
```

custom end marker

- You can choose an end marker for cat with << as is shown in this screenshot.
- This construction is called a here directive and will end the cat command.

```
ubuntu@ubuntu-Inspiron-3558:~/test$ cat >file1 <<eof
> Today is a rainy day!
> I am enjoying shell programming
> eof
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file1
Today is a rainy day!
I am enjoying shell programming
ubuntu@ubuntu-Inspiron-3558:~/test$
```

cat command (to copy file)

- The 'cat' command can be used to copy the content of a file into another file.
- **Syntax:** cat (older file name) > (newer file name)

```
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file1
Today is a rainy day!
I am enjoying shell programming
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file1 > file2
ubuntu@ubuntu-Inspiron-3558:~/test$ ls
f3  f4  file1  file2  fn  test3  test4
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file2
Today is a rainy day!
I am enjoying shell programming
ubuntu@ubuntu-Inspiron-3558:~/test$
```

cat command
(to concatenate files)

- The 'cat' command can be used to concatenate the contents of multiple files in a single new file.
- **Syntax:** cat <filename1> <filename2>....> <newFilename>

```
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file2
Today is a rainy day!
I am enjoying shell programming
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file3
Oh!! cat command is too helpful!!
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file2 file3 > file4
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file4
Today is a rainy day!
I am enjoying shell programming
Oh!! cat command is too helpful!!
ubuntu@ubuntu-Inspiron-3558:~/test$
```

tac

- The 'tac' command is the reverse of the 'cat' command.
- It is also known as 'cat' backward.
- It will display the file content in reverse order. It prints the last line first, then second last and so on.

- **Syntax:**
  tac <file name>

```
ubuntu@ubuntu-Inspiron-3558:~/test$ cat file4
Today is a rainy day!
I am enjoying shell programming
Oh!! cat command is too helpful!!
ubuntu@ubuntu-Inspiron-3558:~/test$ tac file4
Oh!! cat command is too helpful!!
I am enjoying shell programming
Today is a rainy day!
ubuntu@ubuntu-Inspiron-3558:~/test$
```

grep

- ‘grep’ is short for ‘global regular expressions print’.
- Its basic use is to search for a string in a specified file.
- The basic syntax is “grep (search string) (file name)”
- As output, grep will print every line in the text file that features the search string.
- ‘grep’ is a particularly powerful and versatile command and we shall see some more examples.

nl

- **'nl'** is short for 'number lines'.
- The command "nl filename" will display the contents of the file with each line numbered.

```
nandakumar@LAPTOP-3P3I9551:~/linux/classes/class1$ nl data1.txt
     1  alice
     2  anton
     3  ben
     4  bill
     5  charlie
     6  frank
     7  zoe
```

Summary

- In this session we discussed Linux commands based on file and file contents.

Reference

- Cobbaut P. Linux Fundamentals. Samurai Media Limited; 2016.

Objective

- Familiarize Linux Commands –(Part III)
  - I/O Redirection
  - Filter Commands

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Linux Standard Streams: stdin, stdout, and stderr

- Stdin
  - Stands for standard input. It takes text as input.
- Stdout
  - Stands for standard output.
  - The text output of a command is stored in the stdout stream.
- Stderr
  - Stands for standard error.
  - Whenever a command faces an error, the error message is stored in this stream.

Redirecting streams

- stdin (Input Redirection)
  - The '<' symbol is used for input(STDIN) redirection.

1

```
deepa@blackhawk:~/Documents$ cat < states.txt
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Delhi
Kerala
Chennai
Mumbai
```

2

```
deepa@blackhawk:~/Documents$ cat states.txt
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Delhi
Kerala
Chennai
Mumbai
```

Redirecting streams

- stdout (Output Redirection)
  - The **'>'** symbol is used for output (STDOUT) redirection.
  - The **'>>'** symbol is used to append output to an existing file.

1
2
3
4

```
deepa@blackhawk:~/Documents$ echo "Hello, all welcome to this course" > test.txt
deepa@blackhawk:~/Documents$ cat test.txt
Hello, all welcome to this course
deepa@blackhawk:~/Documents$ echo "Computer Systems Essentials" >> test.txt
deepa@blackhawk:~/Documents$ cat test.txt
Hello, all welcome to this course
Computer Systems Essentials
deepa@blackhawk:~/Documents$
```

Redirecting streams

- stderr (Error Redirection)
  - Files are always open, viz., standard input, standard output, standard error.
  - File descriptor, is associated with each of these files.

File	File Descriptor
Standard Input STDIN	0
Standard Output STDOUT	1
Standard Error STDERR	2

  - By default, error stream is displayed on the screen.

Linux Filter Commands

1. grep
2. sort
3. cat
4. wc
5. cut
6. tee
7. tr

grep

- The grep filter searches a file for a particular pattern of characters, and displays all lines that contain that pattern.
- The pattern that is searched in the file is referred to as the regular expression.
- **grep** stands for **g**lobally search for **r**egular **e**xpression and **p**rint out.

Syntax:

- **grep [options] pattern [files]**

Grep

1
```
deepa@blackhawk:~/Documents$ cat datafile.txt
Computer Science is the study of computers and computational systems.
Unlike electrical and computer engineers, computer scientists deal mostly with software and software systems.
This includes their theory, design, development, and application.
```

2
```
deepa@blackhawk:~/Documents$ grep "computer" datafile.txt
Computer Science is the study of computers and computational systems.
Unlike electrical and computer engineers, computer scientists deal mostly with software and software systems.
```

Grep Examples with Options

3
```
deepa@blackhawk:~/Documents$ grep -c "computer" datafile.txt
2
```

4
```
deepa@blackhawk:~/Documents$ grep -w "computer" datafile.txt
Unlike electrical and computer engineers, computer scientists deal mostly with software and software systems.
```

5
```
deepa@blackhawk:~/Documents$ grep -o "computer" datafile.txt
computer
computer
computer
```

6
```
deepa@blackhawk:~/Documents$ grep -n "soft" datafile.txt
2:Unlike electrical and computer engineers, computer scientists deal mostly with software and software systems.
```

7
```
deepa@blackhawk:~/Documents$ grep -i "computer" datafile.txt
Computer Science is the study of computers and computational systems.
Unlike electrical and computer engineers, computer scientists deal mostly with software and software systems.
```

sort

- SORT command is used to sort a file, arranging the records in a particular order.
- By default, the sort command sorts file assuming the contents are ASCII.
- Using options in sort command, it can also be used to sort numerically.

1
2
3
4

```
deepa@blackhawk:~/Documents$ cat  studentlist.txt
Arya
Deepthi
Rahul
Vinay
Bobby
Priya
deepa@blackhawk:~/Documents$ sort studentlist.txt
Arya
Bobby
Deepthi
Priya
Rahul
Vinay
deepa@blackhawk:~/Documents$ cat marks.txt
67
99
76
90
100
40
57
deepa@blackhawk:~/Documents$ sort -n marks.txt
40
57
67
76
90
99
100
```

cat

- This command is used inside pipes, it does nothing except moving (or redirects) stdin to stout.

```
mithun@LAPTOP-G8OP3754:~/study$ cat Professors.txt
159 Alice CSE
196 Charlie CSE
257 Anna CSE
110 Emly CSE
mithun@LAPTOP-G8OP3754:~/study$ cat Professors.txt | sort
110 Emly CSE
159 Alice CSE
196 Charlie CSE
257 Anna CSE
```

wc

- wc stands for word count.
- It is used to find out number of lines, word count, byte and characters count in the files specified in the file arguments.
- By default it displays four-columnar output.
- First column shows number of lines present in a file specified, second column shows number of words present in the file, third column shows number of characters present in file and fourth column itself is the file name which are given as argument.
- Syntax: **wc [OPTION]... [FILE]...**

wc Example

1
```
deepa@blackhawk:~/Documents$ cat datafile.txt
Computer Science is the study of computers and computational systems.
Unlike electrical and computer engineers, computer scientists deal mostly with software and software systems.
This includes their theory, design, development, and application.
```
2
```
deepa@blackhawk:~/Documents$ wc datafile.txt
  3  32 246 datafile.txt
```
3
```
deepa@blackhawk:~/Documents$ wc marks.txt studentlist.txt
 7  7 22 marks.txt
 6  6 37 studentlist.txt
13 13 59 total
```

4
```
deepa@blackhawk:~/Documents$ wc -l datafile.txt
3 datafile.txt
```
5
```
deepa@blackhawk:~/Documents$ wc -w datafile.txt
32 datafile.txt
```
6
```
deepa@blackhawk:~/Documents$ wc -c datafile.txt
246 datafile.txt
```
7
```
deepa@blackhawk:~/Documents$ wc -m datafile.txt
246 datafile.txt
```

cut

- The cut command in UNIX is a command for cutting out the sections from each line of files and writing the result to standard output.
- It can be used to cut parts of a line by byte position, character and field.
- Syntax: **cut OPTION... [FILE]...**

```
deepa@blackhawk:~/Documents$ cat states.txt
Andhra Pradesh
Arunachal Pradesh
Assam
Bihar
Delhi
Kerala
Chennai
Mumbai
deepa@blackhawk:~/Documents$ cut states.txt
cut: you must specify a list of bytes, characters, or fields
Try 'cut --help' for more information.
```

Cut Command with Options

How to cut by byte position

1

```
deepa@blackhawk:~/Documents$ cut -b 3- states.txt
dhra Pradesh
unachal Pradesh
sam
har
lhi
rala
ennai
mbai
```

2

```
deepa@blackhawk:~/Documents$ cut -b -3 states.txt
And
Aru
Ass
Bih
Del
Ker
Che
Mum
```

How to cut by character

3

```
deepa@blackhawk:~/Documents$ cut -c 1,3,5 states.txt
Adr
Aua
Asm
Bhr
Dli
Krl
Cen
Mma
```

4

```
deepa@blackhawk:~/Documents$ cut -c 1-4 states.txt
Andh
Arun
Assa
Biha
Delh
Kera
Chen
Mumb
```

Cut Command with Options

How to cut based on a delimiter

Example 1:

```
deepa@blackhawk:~/Documents$ cat file.txt
245:789 4567 M:4540 Admin 01:10:1980
535:763 4987 M:3476 Sales 11:04:1978
deepa@blackhawk:~/Documents$ cut -d " " -f 2,4 file.txt
4567 Admin
4987 Sales
```

Example 2:

```
deepa@blackhawk:~/Documents$ cat  names.txt
John,Smith,34,London
Arthur,Evans,21,Newport
George,Jones,32,Paris
deepa@blackhawk:~/Documents$ cut -d "," -f 2,4 names.txt
Smith,London
Evans,Newport
Jones,Paris
```

tee

- tee command reads the standard input and writes it to both the standard output and one or more files.
- Syntax:

tee [OPTION]... [FILE]...

tr

- **tr** stands for **translate.**
- The tr is a command line utility for translating or deleting characters.
- It supports a range of transformations including uppercase to lowercase, squeezing repeating characters, deleting specific characters and basic find and replace.
- Syntax : tr [OPTION] SET1 [SET2]

```
1 mithun@blackhawk:~/Documents$ cat test
  hello world
2 mithun@blackhawk:~/Documents$ tr a-z A-Z < test
  HELLO WORLD
3 mithun@blackhawk:~/Documents$ tr -s '[:space:]' '#' < test
  hello#world#mithun@blackhawk:~/Documents$
4 mithun@blackhawk:~/Documents$ cat new
  {computer} {science}
5 mithun@blackhawk:~/Documents$ tr '{}' '()' < new
  (computer) (science)
```

sed

- sed stands for 'stream editor'.
- It parses and transforms text.
- Example : "sed –n 5p filename" will print the 5th line of the file named.
- Note that sed is a utility with great many options and potential.

```
nandakumar@LAPTOP-3P3I9551:~/linux/classes/class1$ cat data1.txt
alice
anton
ben
bill
charlie
frank
zoe
nandakumar@LAPTOP-3P3I9551:~/linux/classes/class1$ sed -n 3p data1.txt
ben
```

Summary

- In this session we discussed Linux commands based on input output redirection commands and filter commands.

Reference

- Cobbaut P. Linux Fundamentals. Samurai Media Limited; 2016.

Objective

- Familiarize Linux Commands –(Part IV)
  - Basic Unix Tools
  - Pipes in Linux

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Basic Unix Tools

- find
- locate
- date
- time

find

- find is a powerful command to search for files and folders in a file hierarchy.
- Example :
  - "find *" will list (with paths) all files and folders in the hierarchy with present directory as root.

```
mithun@LAPTOP-G8OP3754:~/folder2$ find *
demofolder
demofolder/data
file_1
google
ipaddress
ipaddress_1
script1.sh
script2.sh
script3.sh
script4.sh
script5.sh
testfolder
testfolder/text
```

  - "find ." will list (with paths) all files and folders including invisible ones.

```
mithun@LAPTOP-G8OP3754:~/folder2$ find .
.
./demofolder
./demofolder/data
./file_1
./google
./ipaddress
./ipaddress_1
./script1.sh
./script2.sh
./script3.sh
./script4.sh
./script5.sh
./testfolder
./testfolder/text
```

locate

- Used to search a file by name.
- Similar to find command.
- Difference:
  - locate command is a background process and searches the file in the database.
  - find command searches in the filesystem.
- The locate command is much faster than find command.
- Syntax: locate [option] {pattern}

```
mithun@LAPTOP-G8OP3754:~$ locate script5.sh
/home/mithun/folder2/script5.sh
```

date

- Date command is used to display date, time, time zone, etc.

```
mithun@LAPTOP-G8OP3754:~$ date
Tue Mar 30 00:43:39 IST 2021
```

time

- Time command displays how long it takes to execute a command.
- It helps in checking the performance of the scripts and commands.

```
mithun@LAPTOP-G8OP3754:~$ time ls
f1  folder2  google  pingmanual  script6.sh  shell.sh  ssh_keys  study

real    0m0.015s
user    0m0.000s
sys     0m0.000s
```

Pipe in Linux

- A pipe is a form of redirection used to send the output of one command/ program/ process to another command/ program/ process for further processing.

- The Linux systems allow stdout of a command to be connected to stdin of another command.
- It is done by using the pipe character **'|'**.
- Pipes are unidirectional i.e. data flows from left to right through the pipe.
- Syntax :

  command_1 |command_2 |command_3 |.... |command_N

Example 1:

Listing all files and directories and write the output to a new file.

```
deepa@blackhawk:~/Documents$ ls -l | cat > new.txt
deepa@blackhawk:~/Documents$ cat new.txt
total 40
-rw-rw-r-- 1 deepa deepa  246 Jan  7 19:35 datafile.txt
-rw-rw-r-- 1 deepa deepa    0 Jan  7 19:56 errorfile.txt
-rw-rw-r-- 1 deepa deepa    0 Jan  7 19:57 error.log
drwxrwxr-x 2 deepa deepa 4096 Jan  7 19:55 Exam
-rw-rw-r-- 1 deepa deepa   52 Jan  7 19:40 file1.txt
-rw-rw-r-- 1 deepa deepa   74 Jan  8 18:11 file.txt
-rw-rw-r-- 1 deepa deepa   22 Jan  7 20:16 marks.txt
-rw-rw-r-- 1 deepa deepa   67 Jan  8 18:17 names.txt
-rw-rw-r-- 1 deepa deepa  702 Jan  9 02:32 new1.txt
-rw-rw-r-- 1 deepa deepa    0 Jan  9 02:33 new.txt
-rw-rw-r-- 1 deepa deepa   73 Jan  7 20:25 states.txt
-rw-rw-r-- 1 deepa deepa   37 Jan  7 20:12 studentlist.txt
-rw-rw-r-- 1 deepa deepa   62 Jan  7 19:53 test.txt
```

Example 2:

Use head and tail to print lines in a particular range in a file.

```
deepa@blackhawk:~/Documents$ cat poem.txt
You may write me down in history
With your bitter, twisted lies,
You may trod me in the very dirt
But still, like dust, I'll rise.
Does my sassiness upset you?
Why are you beset with gloom?
'Cause I walk like I've got oil wells
Pumping in my living room.
Just like moons and like suns,
With the certainty of tides,
Just like hopes springing high,
Still I'll rise.
```

```
deepa@blackhawk:~/Documents$ cat poem.txt | head -7 | tail -3
Does my sassiness upset you?
Why are you beset with gloom?
'Cause I walk like I've got oil wells
```

Example 3:

Find the no. of "Color LaserJet" and write the details to a new file.

Open ▾ + hp.txt ~/Documents Save

```
HP LaserJet Enterprise M507n 1PV86A#BGJ Black_and_White_Networked updated 20190926
HP LaserJet Enterprise M507dn 1PV87A#BGJ Black_and_White_Networked_with_Duplexing updated 20190926
HP LaserJet Enterprise M607n K0Q14A#BGJ Black_and_White_Networked updated 20171211
HP LaserJet Enterprise M607dn K0Q15A#BGJ Black_and_White_Networked_with_Duplexing updated 20171211
HP LaserJet Enterprise M553n B5L24A#BGJ Color_Networked updated 20151014
HP LaserJet Enterprise M553dn B5L25A#BGJ Color_Networked with Duplexing updated 20151014
HP ColorLaserJet Enterprise M750n D3L08A#BGJ Color_Networked updated 20160902
HP ColorLaserJet Enterprise M750dn D3L08A Color_Networked with Duplexing updated 20160902
```

```
deepa@blackhawk:~/Documents$ cat hp.txt | grep "Color LaserJet" | tee hpcolor.txt | wc -l
2
```

Example 4:

Display the HP printer model and type from the list (hp.txt).

Open + hp.txt ~/Documents Save

```
HP LaserJet Enterprise M507n 1PV86A#BGJ Black_and_White_Networked updated 20190926
HP LaserJet Enterprise M507dn 1PV87A#BGJ Black_and_White_Networked_with_Duplexing updated 20190926
HP LaserJet Enterprise M607n K0Q14A#BGJ Black_and_White_Networked updated 20171211
HP LaserJet Enterprise M607dn K0Q15A#BGJ Black_and_White_Networked_with_Duplexing updated 20171211
HP LaserJet Enterprise M553n B5L24A#BGJ Color_Networked updated 20151014
HP LaserJet Enterprise M553dn B5L25A#BGJ Color_Networked with Duplexing updated 20151014
HP ColorLaserJet Enterprise M750n D3L08A#BGJ Color_Networked updated 20160902
HP ColorLaserJet Enterprise M750dn D3L08A Color_Networked with Duplexing updated 20160902
```

```
deepa@blackhawk:~/Documents$ cat hp.txt | cut -d " " -f 1,2,5,6 | tee updated.txt
HP LaserJet 1PV86A#BGJ Black_and_White_Networked
HP LaserJet 1PV87A#BGJ Black_and_White_Networked_with_Duplexing
HP LaserJet K0Q14A#BGJ Black_and_White_Networked
HP LaserJet K0Q15A#BGJ Black_and_White_Networked_with_Duplexing
HP LaserJet B5L24A#BGJ Color_Networked
HP LaserJet B5L25A#BGJ Color_Networked
HP ColorLaserJet D3L08A#BGJ Color_Networked
HP ColorLaserJet D3L08A Color_Networked_with_Duplexing
deepa@blackhawk:~/Documents$ cat updated.txt
HP LaserJet 1PV86A#BGJ Black_and_White_Networked
HP LaserJet 1PV87A#BGJ Black_and_White_Networked_with_Duplexing
HP LaserJet K0Q14A#BGJ Black_and_White_Networked
HP LaserJet K0Q15A#BGJ Black_and_White_Networked_with_Duplexing
HP LaserJet B5L24A#BGJ Color_Networked
HP LaserJet B5L25A#BGJ Color_Networked
HP ColorLaserJet D3L08A#BGJ Color_Networked
HP ColorLaserJet D3L08A Color_Networked_with_Duplexing
```

Example 5:

Find the name, branch, native place of students from CSE department.

studentlist.txt
~/Documents

```
Arya     CSE       Chennai    46312
Deepthi     ECE    Bangalore      21323
Rahul   CSE    Delhi     56151
Vinay     EEE     Mumbai    244234
Maya    AIE     Bangalore     78699
Priya     CSE     Kerala    510090
```

```
mithun@blackhawk:~/Documents$ cat studentlist.txt|tr -s " " | grep CSE|cut -d " " -f 1-3
Arya CSE Chennai
Rahul CSE Delhi
Priya CSE Kerala
```

Summary

- In this session we discussed Linux commands based on basic Unix tools and pipes in Linux.

Reference

- Cobbaut P. Linux Fundamentals. Samurai Media Limited; 2016.